

Leaf Recognition Algorithm using Support Vector Machine with Hu Moments and Local Binary Patterns

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Abstract—Leaf recognition is convenient for plant classification and it is an important subfield of pattern recognition. Different leaf features such as color, shape and texture are used as well as different classifiers including artificial neural networks, k-nearest neighbor and support vector machines. In this paper we propose an algorithm based on tuned support vector machine as a classifier and Hu moments and uniform local binary pattern histogram parameters as features. Our proposed algorithm was tested on leaf images from standard benchmark database and compared with other approaches from literature where it proved to be more successful (higher recognition percentage).

Index Terms—Leaf recognition, plant classification, support vector machines, Hu moments, local binary patterns.

I. INTRODUCTION

Digital image processing has widespread use in many science disciplines such as astronomy [1], biology [2], medicine [3], food industry [4], criminology [5], etc. Another area of usage of digital image processing is for plant classification. This represents active and important research issue. Due to a huge number of plant species this task has been proven to be very hard. Plants are among the most important life forms on our planet, we use them in medicine, for food, in many different industries, and most importantly plants are the lungs of the Earth. They produce oxygen as a byproduct of photosynthesis, so they are of crucial importance for the survival of life on our planet. Through history humans always had close relationship with plants due to their importance and learned to classify them in many different ways. Nowadays development of technology, especially computer science, created way for classification of plants by use of the computers.

Plant classification is an interesting topic for many different professions related to agriculture, medicine, nature conservation, forestry, furniture industry and in other various situations of general public interest. It is estimated that the number of the plant species is larger than four hundred thousand, many of those are widely used and of great

importance to humanity, so it is clear why there is a need for efficient way for plant classification [6]. Classification can be done by using different plant parts. It was concluded that one of the most suitable ways is by leaf analysis. There are several reasons why leaves are usually chosen for this task: leaves are readily available, they are easily found during most of the seasons, it is easy to collect and preserve them and probably most importantly, leaves have sufficient specificity for almost all species. Also, very important characteristic of the leaves is that they are two-dimensional shapes. It is very hard to use other plant parts (flowers, seeds, trees, etc.) because of their three-dimensionality and often complex structure [7].

Plant classification by leaf image uses digital image algorithms for preprocessing the image and preparing it for the second step: recognition. Object recognition is widely researched topic. Some of the examples are digit recognition [8], [9], text recognition [10], face detection [11], lip recognition [12], etc. Preprocessing of the image contains several steps: image enhancement, feature extraction and segmentation. All of these steps are also part of many other applications, thus they are researched separately. Image enhancement can be done by contrast enhancement [13], [14]. In object recognition, segmentation is a very important step. Numerous methods need just binarization, but often some multilevel thresholding is necessary [15], [16], [17], [18].

Leaves have many characteristics than can be used for classification, some proven to be better than the others. Most commonly used are shape, color and texture [19], but some researchers [20] use more features for more precise methods for leaf classification.

Shape feature usually consist of two geometrical features [21]: aspect ratio (ratio between length and width of the leaf) and compactness or roundness which is ratio between area of the leaf and square of perimeter of the leaf. Other important shape features are: rectangularity or similarity between a leaf and a rectangle, solidity or a ratio between the area of the leaf and the area of its convex hull, area of the leaf witch is the number of pixels that leaf contains. Leaf size has also been examined as one of the features but since true size cannot be

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determined from images taken by camera this feature does not appear to be important. It is important to mention that some plant species have leafs with irregular shapes.

Color features are used by some methods but due to the fact that many plant species have similar leaf colors this feature is not of crucial importance. However, information from histogram can be used for classification of leafs.

Texture features are important as well. They can be used to describe vein structure of the leaf or venation. This feature proved to be very important, right after the shape, for recognition of leafs.

Combination of features mentioned above is usually used to classify leafs and therefore to classify plants. As an input image of the leaf is used while output is the name of the plant species to which leaf belongs.

Recognition can be done by several different techniques: neural networks, support vector machines, nearest neighbor classifier, etc.

In this paper we propose a method for leaves classification. Invariant moments are used as features and principal component analysis was done for feature selection. Classification is done by support vector machine. Support vector machine (SVM) is a discriminative classifier formally defined by a separating hyperplanes. When given labeled training data (supervised learning), the algorithm outputs an optimal hyperplane which categorizes new examples.

This paper is organized as follows. In section II literature review of methods for classifications of plants by leaf images is presented. In section III our methods for classifications of plants by leaf images is proposed. Section IV contains the results obtained by our proposed algorithm. And finally section V gives a conclusion.

II. LITERATURE REVIEW

Many different approaches were used for classification of plants by images of the leafs. Most of them rely on the shape feature of the leafs [22], [23], but some use more features for classification [21]. Almost all approaches have in common the first step which is use of digital image processing for preprocessing the image for classification procedure. Differences come with the second step, classification procedure, where some researchers [20] use neural networks as a tool, while others use support vector machines [24]. Some of the researcher suggest nearest neighbor classifier (1-NN), k-nearest neighbor classifier (k-NN) or combination of both [7].

In paper [25] three techniques for plants classification based on their leaf shape were presented and compared: support vector machine based binary decision tree (SVM-BDT), probabilistic neural network (PNN) and Fourier moment. The proposed SVM-BDT architecture takes advantage of both PNN and Fourier moment technique when it comes to both accuracy and speed of the procedure.

Two techniques were compared in paper [26]: co-occurrence matrix (GLCM) and principal Component analysis (PCA). Both were tested and PCA prevailed with 98 to 78 percents.

Different approach was presented in paper [27]. Authors suggested a novel framework using shape, vein, color and texture features which were combined with Zernike moments. A dual stage training algorithm was used to train RBPNN and method was tested on Flavia data set with high accuracy.

An approach using probabilistic neural network (PNN) was presented in [20]. Unlike many researchers that suggested using of only few leaf features for classification in this paper 12 leaf features were extracted and orthogonalized into 5 principal variables which form the input vector for the PNN. Method had more than 90 percents of success rate. A very different approach was presented in paper [28]. Authors used a two descriptors based on mathematical morphology. First consists of the computation of morphological covariance on the leaf contour profile and the second is an extension of the recently introduced circular covariance histogram, capturing leaf venation characteristics. Approach was tested on ImageClef12 plant identification dataset.

Shape based method that extract the contour signature from every leaf and then calculates the dissimilarities between them using the Jeffrey-divergence measure was presented in paper [22]. Also, they used orientations of edge gradients to analyze macro-texture of the leaf, and then combined two methods using an incremental classification algorithm.

Same group of researchers also suggested shaped based methods in papers [23] and [29], the first one using set of points to describe shape and the second triangle area representation (TAR) to classify leafs.

A new classification method, move median centers (MMC) hypersphere classifier based on digital morphological feature was proposed in paper [30]. Proposed method was proved to be both robust and efficient. Authors used self created data base of leafs.

In paper [31] supervised locally linear embedding (SLLE) was presented as a method for collecting features for simple classifiers. Authors claim that it is a powerful feature extraction method, which gives very promising recognition results.

In [32] a hybrid deep learning that uses auto-encoder and convolutional neural network was proposed. The hybrid deep learning was used for extracting the features of the plant leaves and for classification SVM with extracted features was used.

A unique method for leafs recognition from paper [33] is addressing a problem of two-dimensional (2-D) shape representation and matching in presence of self-intersection for large image databases.

A texture based method based on volumetric fractal dimension was proposed in paper [34]. The proposed method produces a texture signature which is able to efficiently characterize different texture classes. The high classification rate yielded shows that the potential of this method is higher than those which use Gabor filters and Fourier analysis.

A method that rely on probabilistic integration of shape, texture and margin features was suggested in paper [35]. They used the k-Nearest Neighbor (K-NN) classifier as a tool for pattern analysis. Method had a 93 percents of success rate.

Another shape-based method was used in paper [36] that reduces the number of points for the shape representation considerably with definition of new contour descriptor. Method uses two-step algorithm, at the first step the database is reduced by using some geometrical features and at the second step a similarity is measured between the contour representations and used to rank conveniently leaf images from the database.

Elliptic Fourier leaf shape analysis is presented in paper [37]. Chain encoded, elliptic Fourier harmonic functions were generated based on leaf boundary in this paper, then the complexity index of leaf shapes was computed. Plants of different ages were used for experimental results and success rate grow from around 70 percents on younger ones to over 90 percents on older ones.

Most of the mentioned methods use data bases with leafs on simple, often one colored background. In paper [38] a method for classification of plant leaf images with complicated background was presented. As the first step they used automatic marker-controlled watershed segmentation method to extract leaf from image then seven Hu geometric moments and sixteen Zernike moments were used to extract shape features. Method was proven to be efficient with more than 93 percent of success rate.

Although many of the methods were rather successful, there is still place for improvement, especially with raw images so this is still active topic among researchers. Recent development of mobile smart phones with high-resolution cameras opens new opportunities for application that could classify plant from on-line databases just by taking a picture of a plants leaf.

In the next section our proposed method for classification of plants via leaf images is presented.

III. THE PROPOSED METHOD

For high accuracy of classification it is necessary that instances are represented by some invariant features. In image processing, features such as color, shape and texture are common choice for input of the classifier. Even though color features have some advantages such as processing speed and simplicity in many cases they are not suitable. Some of the main issues are color representation as well as light condition at the moment when the picture was taken. Color features can be different just because of technology, i.e. the same object filmed at the same time with different cameras can have distinct color features. Because of these reasons, color features are usually used combined with some other features.

In the case of leaves, shape can be considered as one of the major properties for classification. Nowadays, different shape descriptors are used and all of them can be categorize as contour-based or region-based descriptors. In this paper we use one of the widely used contour-based shape descriptors, regular invariant moments. Invariant moments were introduced by Hu [39]. Today, many different geometrical invariant moments are used such as Chebyshev, Zernike and others. Invariant moments have been widely used in pattern recognition because of their characteristics that can

recognize same objects regardless of their size, position and orientation.

In this paper seven Hu moments are used. Hu moments represent suitable shape features because they are invariant to scaling, translation, rotation and mirroring.

Two dimensional moment of order (p, q) for a digital image $f(x, y)$ of size $M \times N$ is defined by the following equation:

$$m_{p,q} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) x^p y^q \quad (1)$$

where p represents the order of x and q is the order of y .

A central moment is represented similarly as the moments described in Eq. 1. Difference is that instead of the x and y values their differentiates from mean values are used. Central moments are defined as:

$$\mu_{p,q} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) (x - x_{avg})^p (y - y_{avg})^q \quad (2)$$

where x_{avg} and y_{avg} are:

$$x_{avg} = \frac{m_{10}}{m_{00}}, \quad y_{avg} = \frac{m_{01}}{m_{00}}. \quad (3)$$

The normalized moments represents the central moments divided by an appropriate power of m_{00} . Equation for calculation of normalized moments is:

$$\eta_{p,q} = \frac{\mu_{p,q}}{m_{00}^{\frac{p+q}{2}+1}} \quad (4)$$

Central and normalized moments are invariant to scaling and translation so this problem is trivially solved. Problem is to find rotational invariants. As it was mentioned before, Hu proposed moments that are invariant to rotation and mirroring. Hu invariant moments represents linear combinations of the central moments through order three. Definition of seven Hu moments is given by following equations:

$$h_1 = \eta_{20} + \eta_{02} \quad (5)$$

$$h_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2 \quad (6)$$

$$h_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \quad (7)$$

$$h_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \quad (8)$$

$$h_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2 + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03}) \quad (9)$$

$$h_6 = (\eta_{20} - \eta_{02})(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \quad (10)$$

$$h_7 = (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})(3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2) - (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03}) \quad (11)$$

Even though shape features are important, some different classes of leafs have almost the same shape. Example is showed in Fig. 1. Fig. 1(a) is a leaf of *pubescent bamboo* class, while in Fig. 1(b) is a leaf of the class *yew plum pine*. In cases like this shape characteristic will not differentiate classes very good. Because of this, additionally we used texture features. Local binary pattern (LBP) represents simple and powerful texture feature descriptor. Binary patterns are obtained from neighbor pixels. Based on central pixel that is used as threshold value, pixels from 3×3 neighborhood were set to 0 or 1. Obtained values are read in clockwise direction, and it represents local binary pattern. Uniform LBP is one of the most used modification where patterns that contain more than two transitions from 0 to 1 (or 1 to 0) are consider as the same pattern. Number of patterns is reduced from 256 to 59. Further, uniform LBP is transformed into decade number and histogram of occurrence frequencies is made. Next step is to normalize this histogram. In our proposed algorithm mean, standard deviation, energy and entropy are used as input feature for classifier.

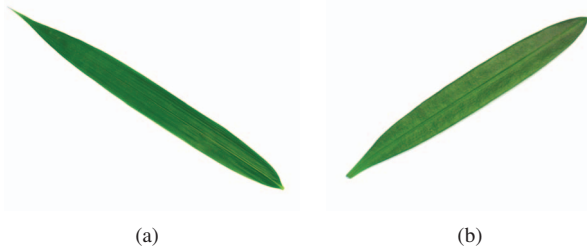


Fig. 1. Example of different classes of leafs with very similar shapes

Hu moments and mentioned LBP histogram parameters were used to describe leaf images. In this paper we used one of the widely used and most successful classifier, support vector machine (SVM) for classification. Support vector machine is a binary classifier. Well known method *one against one* and *one against all* were used in order for binary classifier to perform multi-classification. Quality of the SVM model besides the chosen features, depends on two parameters. Parameters C and γ represent parameters for a nonlinear SVM with a Gaussian radial basis function kernel (RBF). C is the parameter for the soft margin cost function and controls the influence of each individual instance. Kernel function is used for enabling non-linear separable data to be classified by SVM. One of the most used kernel function is RBF which has a free parameter γ . Large γ will lead to high bias and low variance model, and vice-versa. For building a good SVM model, optimal pair (C, γ) need to be found. The simplest method is use of grid search. SVM model is build for numerous pairs with aim to find optimal values for SVM parameters. In our proposed method hierarchical

grid search was used. In the first iteration, parameter C was searched in range $\log_2 C \in \{-5, -3, -1, \dots, 15\}$ and for γ was $\log_2 \gamma \in \{-15, -13, -11, \dots, 5\}$. After finding the best values for (C, γ) , second iteration was done. In the second iteration, search range for each parameter was limited at ± 3 around the previously found best values and the step was reduced to 0.5.

Grid search is computationally expensive, but after the good SVM model is built once, there is no need for building it again. This expensive operation will be done only once, so the price is not so important in our case.

In order to improve performance of our proposed algorithm principal component analysis (PCA) was applied. PCA is well know method used to reduce the dimension of feature vector. This procedure is used on data that contain large number of interrelated variables. Idea is to transform data set into a new set where variables are ordered according to their importance. By applying PCA, components are orthogonalized which uncorrelates them, the result is to sort them in a way that components with the largest variation will appear first and components that have the least contribution will be eliminated. In our case the first component was 6 times larger than the next two components while all others were significantly smaller, at least an order of magnitude. Thus, in this paper we used the first three components of PCA as input features.

IV. EXPERIMENTAL RESULTS

Our proposed algorithm was implemented in Matlab version R2016b and for support vector machine free software LIBSVM (Version 3.21) was used [40]. All experiments were performed on Intel® Core™ i7-3770K CPU at 4GHz, 8GB RAM computer with Windows 10 Professional OS.

There are several databases with leafs that are commonly used for experimental results [19]: Flavia data set, The Swedish data set, ICL (Intelligent Computing Laboratory) data set and ImageCLEF (Cross Language Evaluation Forum) data set. In this paper for the training of SVM and for testing of our proposed algorithm Flavia data set was used. Flavia dataset can be downloaded from <http://flavia.sourceforge.net/>. This dataset contains 1907 images of blades without petioles from 32 different species. Each class has different number of examples. In our experiments 40 instances of each class were used as training set, while accuracy of the obtained model was tested on 10 instances of each class. In Flavia dataset, all images have resolution of 1600x1200. Color leafs are on the white background. Examples of leafs from this dataset are shown in Fig. 2. With our proposed method we achieved accuracy of 94.13%.

In order to evaluate our results we compared accuracy of classification with results of other methods proposed in literature. In [7], first the noise was removed and then the images were segmented. From segmented images the high frequency feature were extracted. For features volume fraction, Hu moments and area convexity were used. Nearest neighborhood classification method was used to classify plant leaf. Proposed method was tested on 320 images of 14 different class of leafs. In [7] recognition accuracy of 91.50%

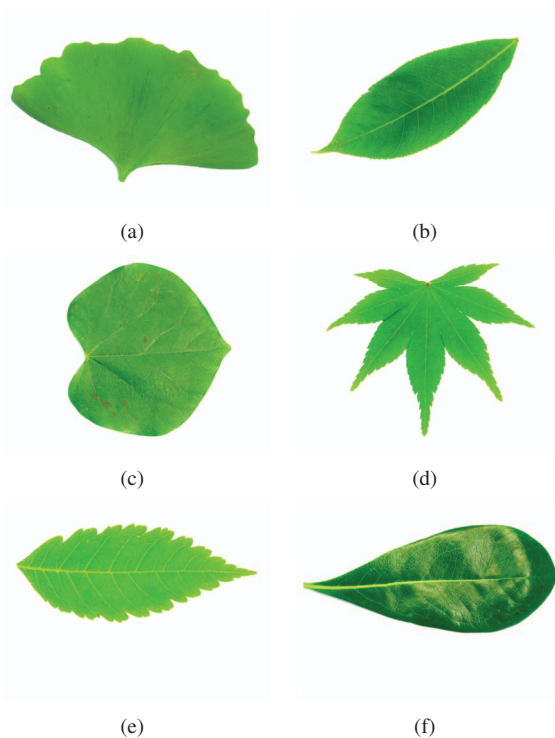


Fig. 2. Examples of leaves from Flavia dataset

V. CONCLUSION

In this paper we described an algorithm for leaf recognition which can be used for plant classification. We used support vector machine as a classifier and tuned it using hierarchical grid search. Invariant Hu moments are very good and discriminative features but inadequate for some cases when leaves from different species have very similar shape. To resolve these cases we additionally used uniform local binary pattern histogram parameters, namely mean, standard deviation, energy and entropy. We tested our proposed algorithm on standard benchmark Flavia data set and achieved accuracy of 94.13 % which is higher compared to other approaches from literature. Future research can improve tuning of the support vector machine by swarm intelligence algorithms, additionally improve features selection and test the algorithm on other benchmark databases.

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was reported. Our results are not completely comparable, since in [7] different images were used and less classes needed to be recognized. Moreover, in [7] classification task is easier, thus it would be expected that higher accuracy was achieved. Since our proposed method obtained better accuracy (94.13% compared to 91.50%) it can be concluded that our proposed method outperformed method proposed in [7].

In [21] for classification of leaves from Flavia dataset, shape, vein, color and texture features were incorporated. In the mentioned paper a probabilistic neural network (PNN) was used as a classifier. The experimental result showed that the best accuracy of 93.75% was achieved when polar Fourier transform (PFT), 3 geometric features, mean, standard deviation and skewness of colors, 12 texture features and 3 vein features were used. Our proposed method achieved accuracy of 94.13%. Besides the fact that our proposed method obtained better global accuracy, number of features is significantly reduced. In our method, only Hu moments and four characteristics of uniform LBP were used which later was reduced by PCA to only 3 features. In [21] accuracy lower than 90% was reported when less features were used. When PFT, 3 geometric features and 12 texture feature were used, accuracy of 85.31% was reported in [21]. Accuracy of classification highly depends on chosen features that describe images, but also on chosen classifier. In this paper we have also shown that support vector machine with tuned parameters is a very good choice.

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